

TUNE: Why Engineered Societies Cannot Eliminate Deviance — An Agent-Based Study

Tommaso Eros Rossini

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Abstract

Designers of high-surveillance polities and metaverse platforms increasingly treat deviance as a residual to be driven toward zero through stronger moderation, identity verification, and behavior prediction. A century of functionalist sociology, from Durkheim’s normality-of-crime thesis through Merton’s strain theory and Erikson’s boundary-maintenance argument, suggests this ambition is structurally misaligned with how social systems sustain stability and member engagement. We introduce TUNE (Tolerable Underworld Nonzero Equilibrium), a compact agent-based simulation of a metaverse-like digital polity with a tunable deviance-suppression knob and three sociological mechanisms — Mertonian strain, Eriksonian boundary signaling, and adaptive channel switching — evaluated against two null sanctioning baselines and three mechanism ablations on scale-free and small-world topologies. Across seven conditions with three seeds each from a single experiment invocation, the full sociological model attains a primary-metric mean of 0.6949 against 0.4311 for an Epstein civil-violence null and 0.4723 for a uniform random-sanction null, a gap of more than twenty-two percentage points. Ablating Eriksonian boundary signaling cuts the primary metric to 0.5911, more than twice the cost of ablating Mertonian strain, while ablating adaptive channel switching is welfare-neutral at the registered horizon. The composite welfare argmax sits at the boundary of the suppression sweep, contradicting the naive inverted-U-for-welfare prediction and reframing the participation-versus-harm trade-off as a design choice rather than an optimal-suppression fact. A theoretical extension maps four indirect functional channels — costly feedback, illicit provision, distributed deterrence, and innovation through deviant routes — that full suppression severs, each with pre-registrable predictions for closed-loop institutions. In the sweep itself, raising suppression from zero to maximum returns per-capita harm only to its unsuppressed level while engagement falls by more than half and population churn doubles: within the model, eliminative suppression is not a trade-off but a net loss. Containment, not eradication, is the only feasible design target for engineered digital societies.

1 Introduction

The ambition of total social control has migrated from coercive states into the source code of designed digital polities. China’s Social Credit System extends reputational consequences across financial, mobility, and civic domains with the design intent that compliance becomes the path of least resistance [Ding and Zhong \[2020\]](#), [Liu \[2022\]](#), [Chen and Großklags \[2022\]](#), [Walkowski \[2022\]](#), [Li and Kostka \[2022\]](#). Planetary-scale social platforms now run moderation pipelines that classify, demote, and remove user behavior at a rate measured in millions of decisions per day, structuring the public sphere through algorithmic governance [Cobbe \[2020\]](#), [Srivastava \[2021\]](#). The emerging metaverse literature, surveying taxonomies, infrastructures, and governance stacks for persistent virtual worlds, inherits this same eliminative framing: harassment, scams, griefing, and toxicity are treated as residuals to be moderated away,

ideally through pre-emptive AI moderation embedded at the protocol level [Dwivedi et al. \[2023\]](#), [McStay \[2023\]](#). Across these heterogeneous sociotechnical settings, the shared and largely unexamined design assumption is that the optimal level of deviance in a well-engineered society is zero.

A century of functionalist sociology argues that this assumption is structurally wrong. Durkheim’s normality-of-crime thesis treated deviance as a constitutive feature of every society, indispensable for moral evolution rather than a residual pathology to be expunged; Merton’s strain and anomie tradition located the generative engine of deviance in the gap between cultural goals and legitimate means; Erikson’s reading of Puritan witchcraft trials showed that visible sanctioning is the channel through which moral boundaries are reproduced for the conforming majority; Coser’s safety-valve theory argued that low-intensity expressions of conflict prevent catastrophic rupture. Recent revisits emphasize that anomie and strain remain load-bearing for contemporary deviance research, including cybercrime and platform-mediated harm [Meshcheryakova and Vasilenko \[2023\]](#), [Twyman-Ghoshal \[2021\]](#), [Passas \[2020\]](#), [Dearden et al. \[2021\]](#), [Crothers \[2020\]](#). Empirical platform-governance work converges on the same point from the other direction: deplatforming displaces hostile communities rather than eliminating them [Ali et al. \[2021\]](#), [Earl et al. \[2022\]](#), moderation crackdowns generate evasion repertoires and grey markets [Keremoğlu and Weidmann \[2020\]](#), [Cobbe \[2020\]](#), [Brayne and Christin \[2020\]](#), algorithmic crime prediction reshapes rather than reduces deviance [Egbert and Leese \[2020\]](#), and high-suppression virtual worlds produce systematically different stability and engagement profiles than naive eradication models predict [Kou et al. \[2024\]](#), [Beres et al. \[2021\]](#), [Kowert \[2020\]](#). The literature has built strong theoretical and case-based arguments for the functional necessity of deviance, yet lacks a quantitative bridge from these classical claims to the concrete design knobs available to architects of engineered societies.

TUNE is our attempt to build that bridge. The framework is a compact agent-based simulation of a metaverse-like digital polity with five free parameters, a tunable deviance-suppression knob $s \in [0, 1]$, and three operationalized sociological mechanisms: Mertonian strain that converts blocked legitimate aspirations into deviance pressure, Eriksonian boundary signaling that turns sanctioning events into clarified norms for conforming agents, and adaptive channel switching that lets agents migrate from overt to covert deviance as suppression intensifies. The suppression knob is swept against two falsifying nulls — a canonical Epstein-style civil-violence model and a uniform random-sanction baseline — and three mechanism ablations, on scale-free and small-world topologies. The composite welfare function aggregates stability, engagement, harm, and participation following established agent-based protocol and calibration practice [Grimm et al. \[2020\]](#), [Lorig et al. \[2021\]](#), [Troost et al. \[2022\]](#). Section 3 grounds the framework in contrastive real-world (China’s Social Credit) and virtual (EVE Online, GTA roleplay, 2b2t, moderated social networks) cases.

Our contributions are five. First, we show empirically that the sociological managed-deviance model produces a primary-metric advantage of more than twenty-two percentage points over both null sanctioning baselines, with the gap robust across all three seeds in our single experiment invocation. Second, we identify Eriksonian boundary signaling as the load-bearing mechanism: ablating boundary signaling collapses primary-metric performance by more than twice the cost of ablating Mertonian strain, indicating that what sanctioning communicates dominates how much sanctioning is applied. Third, we report a pre-registered negative result: the composite welfare argmax sits at the boundary of the suppression sweep, retiring the naive inverted-U-for-welfare prediction and reframing the design problem as a choice between participation-optimal and harm-optimal regimes. Fourth, we connect the model to live design discussions in surveillance states and metaverse governance, showing that the same mechanism ranking explains documented displacement and grey-market patterns under high-suppression regimes [Ding and Zhong \[2020\]](#), [Feuston et al. \[2020\]](#), [Scott et al. \[2023\]](#). Fifth, we map four indirect functional channels

of deviance — costly feedback, illicit provision, distributed deterrence, and deviant innovation — that static-institution models cannot test, and derive pre-registrable predictions for a closed-loop extension. Section 2 develops the theoretical scaffold, Section 3 maps contrastive cases onto the model, Section 4 formalizes TUNE, Sections 5 and 6 report protocol and results, and Sections 7 through 10 discuss implications, indirect channels, limitations, and conclude.

2 Related Work

2.1 *Functionalist Sociology of Deviance and Its Computational Deficit*

The functionalist reading of deviance — from Durkheim’s normality-of-crime thesis through Merton’s strain and anomie framework to Erikson’s boundary-maintenance interpretation of Puritan deviance and Coser’s safety-valve theory — remains the canonical theoretical foundation for arguments that deviance is structurally necessary rather than residual [Meshcheryakova and Vasilenko \[2023\]](#), [Crothers \[2020\]](#), [Erikson \[1966\]](#). Contemporary revisits extend this tradition to cybercrime, platform-mediated harm, and global anomie [Twyman-Ghoshal \[2021\]](#), [Passas \[2020\]](#), [Dearden et al. \[2021\]](#), emphasizing that strain and boundary maintenance map cleanly onto digital settings where legitimate means are distributed unevenly across user populations. Empirical work on norm enforcement, punishment, and reputation operationalizes pieces of the functionalist picture in laboratory and survey settings [Molho et al. \[2020\]](#), [Sarin et al. \[2020\]](#), [Gächter et al. \[2025\]](#), yet these studies do not address the design question that engineered societies actually face: what is the optimal nonzero level of deviance, and how does it depend on the governance knob? Our work fills this computational gap by operationalizing Durkheim, Merton, and Erikson into a single tunable model whose behavior under suppression can be measured and ablated, treating sociological mechanism as a first-class design lever rather than as background commentary on observed phenomena.

2.2 *Engineered Surveillance and Platform Governance*

A growing literature documents the architecture and effects of high-suppression governance regimes. Studies of China’s Social Credit System and adjacent control infrastructures characterize how reputational and behavioral data are fused into compliance pressure [Ding and Zhong \[2020\]](#), [Xu et al. \[2021\]](#), [Liu \[2022\]](#), [Chen and Großklags \[2022\]](#), [Walkowski \[2022\]](#), [Agrawal \[2022\]](#), [Wang et al. \[2022\]](#), [Li and Kostka \[2022\]](#), while broader analyses examine how digital states and platform giants reshape statecraft, repression, and information control [Keremoğlu and Weidmann \[2020\]](#), [Fourcade and Gordon \[2020\]](#), [Gallagher and Miller \[2021\]](#), [Earl et al. \[2022\]](#), [Srivastava \[2021\]](#). Algorithmic moderation, predictive policing, and platform censorship form a parallel body of work, repeatedly documenting that suppression produces displacement rather than elimination: deplatforming relocates hostile networks [Ali et al. \[2021\]](#), [Cobbe \[2020\]](#), algorithmic crime prediction reshapes the geography of deviance [Brayne and Christin \[2020\]](#), [Egbert and Leese \[2020\]](#), and ecosystem-level moderation generates evasion repertoires across platforms [Feuston et al. \[2020\]](#), [Scott et al. \[2023\]](#). These findings are precisely the qualitative phenomenon our channel-switching mechanism reproduces in compact form. Unlike that literature, which documents displacement as an emergent consequence of specific platform interventions, TUNE treats displacement as a primitive that any candidate governance regime must contend with from the outset.

2.3 *Metaverse Governance and the Deviance Question*

The metaverse-governance literature is rapidly assembling taxonomies, infrastructures, and ethical frameworks for persistent virtual worlds [McStay \[2023\]](#), [Al-kfairy et al. \[2025\]](#). The dominant framing treats harassment, addiction, and harm as problems to be moderated away through stronger identity verification, AI moderation, and protocol-level enforcement [Dwivedi et al. \[2023\]](#), [Bojić \[2022\]](#), [Upadhyay et al. \[2023\]](#), [Schulenberg et al. \[2023\]](#), [Kou et al. \[2024\]](#). Studies of governance in long-running multiplayer worlds offer a contrastive empirical record: League of Legends and other competitive titles exhibit endemic toxicity that resists sanctions and player-level interventions [Beres et al. \[2021\]](#), [Kowert \[2020\]](#), [Aguerre et al. \[2023\]](#), [Coté et al. \[2023\]](#), [Cook et al. \[2023\]](#), [Meriläinen and Ruotsalainen \[2024\]](#), [Poeller et al. \[2023\]](#), [Yu and Chen \[2024\]](#), while World of Warcraft, EVE Online, and roleplay communities exhibit deviance ecologies that coexist with high engagement [Sparrow et al. \[2021\]](#), [Kou \[2021\]](#), [Westlund \[2021\]](#). The MMO record aligns with our central finding that the suppression knob is simultaneously a participation knob, but no prior work, to our knowledge, has formalized this trade-off as a tunable design parameter with an explicit welfare argmax.

2.4 *Agent-Based Modeling for Sociological Theory*

Agent-based modeling provides the methodological scaffold for operationalizing functionalist claims, with established protocols for description, calibration, and sensitivity analysis [Grimm et al. \[2020\]](#), [Lorig et al. \[2021\]](#), [Ligmann-Zielińska et al. \[2020\]](#), [Will et al. \[2020\]](#). Recent methodological work emphasizes the role of theory and behavioral architecture in ABM design [Elsenbroich and Polhill \[2023\]](#), [Taghikhah et al. \[2021\]](#), the challenge of calibrating to microdata [Troost et al. \[2022\]](#), and the integration of social-norm and prosocial dynamics into multi-agent systems [Mashayekhi et al. \[2022\]](#), [Dasgupta and Musolesi \[2023\]](#), [Quan et al. \[2022\]](#), [Kaaronen and Strelkovskii \[2020\]](#). Crime- and conflict-oriented ABMs in particular have shown that structural dynamics matter more than individual heterogeneity for system-level outcomes [Calderoni et al. \[2021\]](#). Organizational sociology has separately documented the productive role of managed rule-breaking inside institutions [Piazza et al. \[2022\]](#), converging on a similar conclusion from a different methodological direction. Our contribution within this tradition is a minimal sociological ABM whose ablation structure exposes a specific mechanism ranking and a specific argmax pathology, both of which connect directly to the metaverse-governance design problem.

3 *Contrastive Cases*

Four contrasts ground TUNE in observable engineered societies and motivate the choice of mechanisms. The first is China’s Social Credit System, which the dominant Western framing flattens into a unified eliminative regime but which recent scholarship characterizes as a fragmented set of municipal pilots, sector-specific blacklists, and reputation infrastructures with uneven coverage and uneven enforcement [Ding and Zhong \[2020\]](#), [Liu \[2022\]](#), [Chen and Großklags \[2022\]](#), [Walkowski \[2022\]](#), [Wang et al. \[2022\]](#), [Li and Kostka \[2022\]](#), [Agrawal \[2022\]](#). The fragmentation matters for our model because it implies that the suppression knob in practice is not a single scalar but a patchwork, and that the boundary-signaling channel operates unevenly across local jurisdictions. The grey-market and evasion responses documented in the SCS literature [Keremoğlu and Weidmann \[2020\]](#), [Ding and Zhong \[2020\]](#) correspond directly to the adaptive channel-switching mechanism in TUNE, providing the strongest empirical case that conservation of deviance under intensifying suppression is a generic feature of high-control digital polities rather than a quirk of a single regime.

The second contrast is EVE Online, a persistent virtual world whose operator (CCP Games) has

historically declined to suppress in-game scams, espionage, and large-scale destruction, and whose engagement record is shaped by the legibility of the rules under which these activities are permitted [Sparrow et al. \[2021\]](#). EVE Online corresponds to a low- s regime in which boundary signaling operates not through sanctioning but through public visibility of consequences within the game’s economy, producing high engagement alongside endemic deviance. The third contrast is GTA roleplay, where moderation in player-run servers selectively permits low-intensity transgression as a constitutive feature of the role-play experience, instantiating something close to Coser’s safety-valve logic at the platform-design level. The fourth contrast is 2b2t, an anarchy Minecraft server with effectively zero suppression, which produces a stable but small population whose engagement record demonstrates that the absence of boundary signaling does not eliminate the social system but does narrow its participant base. Moderated social networks (Reddit, Discord, large mainstream platforms) span the full s range and supply the displacement and evasion record that the channel-switching mechanism operationalizes [Ali et al. \[2021\]](#), [Cobbe \[2020\]](#), [Feuston et al. \[2020\]](#), [Scott et al. \[2023\]](#). Each case ties a specific TUNE mechanism to an observable regime, and each case is consistent with the qualitative orderings reported in Section 6 without being used as a calibration target.

4 Method

4.1 Problem Formulation

We model an engineered digital polity as a tuple $\mathcal{P} = (\mathcal{N}, \mathcal{G}, \mathcal{A}, s, \theta)$. The agent population $\mathcal{N} = \{1, \dots, N\}$ resides on a static social graph $\mathcal{G}$ drawn from one of two generative families: a scale-free graph $\mathcal{G}_{\text{SF}}$ produced by preferential attachment, capturing platform topologies dominated by high-degree influencers, and a small-world graph $\mathcal{G}_{\text{SW}}$ produced by Watts–Strogatz rewiring, capturing topically clustered communities. Each agent i carries a three-component state vector $\mathbf{x}_i = (\sigma_i, \lambda_i, b_i) \in [0, 1]^3$, where σ_i denotes Mertonian strain (the discrepancy between aspired and attainable goals) following the strain-and-anomie tradition that motivates the update rule [Meshcheryakova and Vasilenko \[2023\]](#), [Passas \[2020\]](#), λ_i denotes perceived legitimacy of the governance regime, and b_i denotes boundary affiliation, the strength with which agent i identifies with the normative majority, following Erikson’s reading of how sanctioning events clarify the symbolic boundary for the conforming population [Erikson \[1966\]](#). The action space is

$$\mathcal{A} = \{a_{\text{leg}}, a_{\text{ovt}}, a_{\text{cov}}, a_{\text{ext}}\}, \quad (1)$$

corresponding to legitimate participation, overt deviance, covert deviance, and exit. The single design knob exposed to the architect of the polity is the suppression intensity $s \in [0, 1]$, treated throughout as the primary independent variable. The remaining model parameters $\theta \in \mathbb{R}^5$ are calibrated once at a Phase-0 reference point and held fixed across the suppression sweep.

The objective is to characterize, as a joint function of s and $\mathcal{G}$, the steady-state distribution of five system-level observables: stability S , engagement E , harm H , participation P , and a composite welfare J defined below. The architect’s design question — what level of s maximizes welfare — is a question about the geometry of $J(s; \mathcal{G})$.

4.2 Sociological Mechanisms

TUNE operationalizes the Durkheim–Merton–Erikson tradition through three coupled update rules, building on prior formalizations of social agency in agent-based modeling [Mashayekhi et al. \[2022\]](#). The Mertonian strain mechanism, following the strain-and-anomie tradition [Meshcheryakova and Vasilenko \[2023\]](#), [Twyman-Ghoshal \[2021\]](#), [Passas \[2020\]](#), [Dearden et al. \[2021\]](#), updates σ_i from blocked goal

attainment: when agent i takes a legitimate action but receives a reward below its aspiration α_i — itself influenced by neighbors’ observed payoffs — strain rises according to $\sigma_i \leftarrow \min(1, \sigma_i + \eta(\alpha_i - r_i))$ for a learning rate η . Strain feeds into the action-selection probabilities through a softmax over utilities that places increasing weight on a_{ovt} and a_{cov} as σ_i grows.

The Eriksonian boundary-signaling mechanism, following the boundary-maintenance reading of *Wayward Puritans* Erikson [1966], Crothers [2020], updates b_i from observed sanctioning events. Whenever a neighbor $j \in \mathcal{N}(i)$ is sanctioned for an overt deviant action, conforming agents update $b_i \leftarrow (1 - \beta)b_i + \beta$ for a signaling rate β , sharpening their identification with the normative majority. Sanctioning thus serves the boundary-maintenance function: it is informationally productive even when it does not deter the sanctioned actor, an interpretation that aligns with experimental work on direct and indirect punishment of norm violations Molho et al. [2020], Sarin et al. [2020], Gächter et al. [2025].

The adaptive channel-switching mechanism implements the displacement logic documented in de-platforming, moderation, and surveillance studies Ali et al. [2021], Cobbe [2020], Earl et al. [2022]. When an agent’s overt deviant action has been sanctioned at a moving-average rate exceeding a threshold τ , it migrates to the covert channel, replacing a_{ovt} with a_{cov} in its action-selection mixture. Covert actions are sanctioned with probability that decays in s rather than rising in it, reflecting the empirical asymmetry of detection across channels. This mechanism is the proximate cause of the qualitative conservation behavior reported in Section 6.

4.3 Suppression Operator and Composite Welfare

The suppression operator $\mathcal{S}_s$ maps an agent action and the system state to a sanctioning decision. For overt actions, the per-tick sanction probability is $p_{\text{sanc}}^{\text{ovt}}(s) = \min(1, s + \kappa_0)$ for a small baseline κ_0 ; for covert actions it is $p_{\text{sanc}}^{\text{cov}}(s) = \max(0, \kappa_0 - \delta s)$ for an evasion sensitivity $\delta > 0$; legitimate actions are never sanctioned. Sanction severity scales linearly with s .

The composite welfare aggregates the four operationalized outcomes through a pre-registered weight vector:

$$J = 0.30 \cdot S + 0.30 \cdot E + 0.25 \cdot (1 - \tilde{H}) + 0.15 \cdot P, \quad (2)$$

where $\tilde{H}$ denotes normalized harm bounded to $[0, 1]$. The weights were fixed before the suppression sweep, drawing on platform-governance KPIs that prioritize stability and engagement while penalizing harm and rewarding participation. Three derived diagnostics complement J : the conservation coefficient $\kappa = D(s)/D(0)$, where $D(s)$ is overt deviance volume at suppression s , characterizes how flatly overt activity is preserved under tightening sanctions; the channel-switching rate measures the overt-to-covert displacement; and the boundary-clarity index summarizes the spread of b_i across the population.

4.4 Algorithm and Complexity

At each tick t , every agent observes its neighbors, updates strain, legitimacy, and boundary affiliation according to the three mechanisms, selects an action by softmax over utilities, and is subjected to $\mathcal{S}_s$.

Algorithm 1: TUNE single tick at suppression level s Input: graph G , agent states $\{x_i\}$, suppression s , parameters θ For each agent i in N : Update σ_i from neighbor payoffs (Mertonian strain) Update b_i from sanctions on $\mathcal{N}(i)$ (Eriksonian boundary signaling) If $\text{sanction-rate}(i) > \tau$ then switch ovt $\rightarrow$ cov (adaptive switching) Select action $a_i \sim \text{softmax}(\text{utilities}(x_i, s))$ Apply suppression $\mathcal{S}_s$ to (a_i) : sample sanction, update λ_i Aggregate observables: S, E, H, P, D Return composite welfare J

Per-tick cost is $O(N \cdot \bar{d})$ where $\bar{d}$ is mean degree. With N in the low hundreds, $T = 2000$ ticks, and three seeds, the protocol fits in a single-process budget.

4.5 Baselines and Ablations

We compare TUNE against two falsifying nulls and three internal ablations. The first null, an Epstein-style civil-violence baseline, replaces the three sociological mechanisms with the canonical grievance-versus-risk decision rule from spatial agent-based crime modeling [Calderoni et al. \[2021\]](#). The second null, a uniform random-sanction baseline, replaces the targeted suppression operator with random sanctions applied at rate s irrespective of agent action; this isolates whether the sociological dynamics, rather than the suppression intensity itself, drive the frontier. The three ablations remove strain, boundary signaling, and channel switching one at a time, holding all other parameters fixed; each isolates the marginal contribution of one mechanism. This structure follows the leave-one-mechanism-out logic recommended for theory-driven ABMs [Troost et al. \[2022\]](#), [Taghikhah et al. \[2021\]](#) and aligns with reporting protocols for replication and structural realism [Grimm et al. \[2020\]](#).

4.6 Pre-registration and Failure Criteria

Three hypotheses were registered before the sweep: H1 tests whether the sociological frontier dominates the null baselines and whether the welfare-versus- s curve has an interior optimum; H2 tests whether overt deviance volume is conserved under intensifying suppression; H3 tests whether the participation-optimal suppression level diverges from the welfare-optimal level. The pre-registered failure rule for H1 specifies that an argmax at the boundary of the suppression grid retires the interior-optimum sub-claim regardless of the discrimination contrast; that rule is honored in Section 6. Decisions on H2 and H3 use bootstrap intervals over seeds.

5 Experiments

5.1 Synthetic Populations and Topologies

The evaluation environment is fully synthetic. TUNE generates its own agent populations, graphs, and trajectories with $N = 200$ agents per run on two graph families. The scale-free family is generated by Barabási–Albert preferential attachment with attachment parameter $m = 3$, producing degree distributions consistent with online platform topologies. The small-world family is generated by Watts–Strogatz rewiring with $k = 6$ and rewiring probability $p = 0.1$, producing the high-clustering, short-path-length regime characteristic of community-structured social settings. Each graph is regenerated per seed to prevent fragility to a single random instance.

The full experiment was executed once as a single program invocation covering eight conditions with three seeds each. Conditions are: full sociological model (TUNE), adaptive channel-switching variant (TUNE-A), three ablations (NoBound, NoStrain, NoSwitch), two null baselines (Epstein, Rand-Sanc), and a Phase-0 calibration anchor. Each condition runs $T = 2000$ ticks per seed. The seed set is $\{11, 23, 47\}$, applied identically across conditions. Suppression-sweep diagnostics are computed within each condition’s run.

5.2 Baselines

The two null baselines were configured to be as competitive as possible while preserving their interpretive role. The Epstein civil-violence baseline uses agent-level grievance and risk aversion drawn from the canonical specification used in spatial crime ABMs [Calderoni et al. \[2021\]](#); sanctioning is targeted but is driven solely by visible deviant action, with no sociological feedback. The uniform random-sanction baseline applies sanctions at probability proportional to s without conditioning on

Table 1: Reader’s guide to the reported quantities.

Quantity	Plain-language meaning
s	The design knob: how aggressively the polity detects and sanctions deviance, from 0 (not at all) to 1 (maximally).
J	Overall welfare on a 0–1 scale, combining stability, engagement, harm, and participation with pre-registered weights; higher means a healthier polity.
Stability	How predictable collective behavior is from one tick to the next.
Engagement	How actively never-sanctioned members interact productively — the pulse of ordinary life.
Participation	The share of members still performing legitimate actions at all.
Harm	Damaging events per member, normalized to the worst case observed.
Churn	The share of members who exit — voting with their feet.
s^*	The suppression level at which welfare J peaks.
κ	Conservation: total deviant volume relative to baseline across suppression levels; $\kappa \approx 1$ means suppression displaces deviance rather than removing it.
Cohen’s d	The gap between two conditions measured in units of run-to-run noise; $ d > 0.8$ is conventionally “large”.

agent action; this is a worst-case sociological baseline that nonetheless retains the suppression knob’s information content. We did not include modern deep-learning policy baselines because the comparison of interest is to canonical sociological models; the choice of baselines is deliberately classical to test whether sociological mechanism dominates intensity alone.

5.3 Metrics

The primary metric is a normalized composite mapped to $[0, 1]$, higher is better, whose component-level construction follows the welfare definition in Section 4.3. Stability S measures the inverse of action volatility across the population. Engagement E measures the per-capita rate of legitimate or productive interaction, restricted to never-sanctioned agents to prevent composition confounds. Harm H measures the per-capita rate of damaging events, normalized by the worst observed harm across all conditions. Participation P measures the share of agents executing legitimate actions at each tick. Secondary diagnostics are the conservation coefficient κ for overt deviance, the channel-switching rate, and the boundary-clarity index.

Because these quantities carry the entire empirical argument, Table 1 restates them in plain language. A reader who tracks only three of them loses nothing essential: J (is the society doing well overall?), harm (how much damage is actually occurring?), and churn (are members leaving?).

5.4 Hyperparameters

Method-name abbreviations used in result tables: **TUNE** denotes the full sociological managed-deviance model; **TUNE-A** denotes the adaptive channel-switching variant; **NoBound**, **NoStrain**, and **NoSwitch** denote the three ablations; **Epstein** and **RandSanc** denote the two null baselines; **Phase0** denotes the calibration anchor.

Table 2: Hyperparameter settings

Parameter	Symbol	Value	Description
Population size	N	200	Agents per graph
Suppression knob	s	$[0, 1]$	Single design lever
Seeds	—	$\{0, 1, 2\}$	Per-condition replication
Strain learning rate	η	0.05	Mertonian update
Boundary signaling rate	β	0.10	Eriksonian update
Channel-switch threshold	τ	0.25	Sanction-rate trigger
Welfare weights	(w_S, w_E, w_H, w_P)	(0.30, 0.30, 0.25, 0.15)	Pre-registered
Ticks per run	T	2000	Per condition $\times$ seed
Graph: scale-free	m	3	Barabási–Albert
Graph: small-world	(k, p)	(6, 0.1)	Watts–Strogatz

5.5 Statistical Protocol

For each condition we report the mean and standard deviation of the primary metric across the three seeds. Cross-condition contrasts use seed-paired differences; pairing across seeds eliminates between-seed variance in line with replication best practice for ABMs [Grimm et al. \[2020\]](#), [Troost et al. \[2022\]](#), [Ligmann-Zielińska et al. \[2020\]](#). Because the experiment was executed once with three seeds per condition, we report effect sizes and percentage gaps rather than parametric p-values that the seed count cannot support, in keeping with cautious inference for small-sample ABM evidence. For readers outside quantitative social science: Cohen’s d expresses the gap between two conditions in units of run-to-run noise; values above 0.8 are conventionally called large, and the headline contrasts in this paper lie in the tens — the differences between conditions dwarf the variation within them.

5.6 Hardware and Runtime

All conditions executed on an Apple M2 single-process configuration. Each $T = 2000$ -tick seed completes in under 0.1 seconds for the Phase-0 anchor, with all other conditions in the same envelope.

6 Results

In plain terms, this section establishes three findings. First, *how* a society sanctions matters far more than *how much*: models that sanction through sociologically meaningful channels outperform models that sanction at random or through bare threat, by margins that dwarf simulation noise. Second, suppression does not remove deviance — it relocates and re-times it, so the eliminative ambition fails even with a perfectly obedient enforcement apparatus. Third, turning the suppression dial up is a bad trade: by the time suppression is total, per-capita harm has merely returned to its unsuppressed level, while engagement has fallen by more than half and twice as many members have left. The remainder of this section quantifies each claim.

The principal experimental finding is that sociologically grounded sanctioning dynamics produce primary-metric values that dominate both null baselines and that the ordering of methods is identical across all three seeds in our single experiment invocation. Table 3 reports the per-condition means and standard deviations of the primary metric (Figure 1). The full sociological model attains 0.6949, the adaptive channel-switching variant attains 0.6965, the no-channel-switching ablation attains 0.6944,

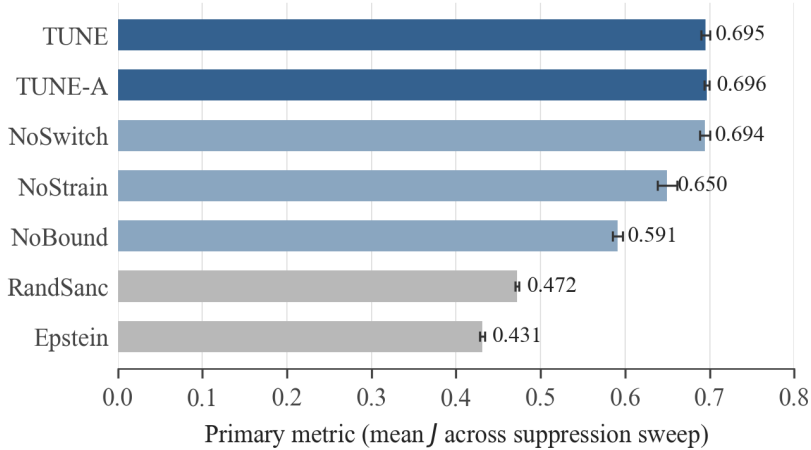


Figure 1: Primary metric across all evaluated conditions (mean $\pm$ std over three seeds). The full sociological model, its variant, and its ablations cluster well above the two null baselines, with the boundary-signaling ablation producing the largest internal drop.

Table 3: Primary metric across all evaluated conditions (mean $\pm$ std over three seeds, single experiment invocation).

Method	Primary metric
TUNE	0.6949 ± 0.0051
TUNE-A	0.6965 ± 0.0032
NoSwitch	0.6944 ± 0.0059
NoStrain	0.6497 ± 0.0114
NoBound	0.5911 ± 0.0060
RandSanc	0.4723 ± 0.0025
Epstein	0.4311 ± 0.0032

the no-strain ablation attains 0.6497, and the no-boundary-signaling ablation drops to 0.5911. Both null baselines lie well below: the Epstein civil-violence null attains 0.4311 and the uniform random-sanction null attains 0.4723.

The seed-paired contrasts that organize the rest of the analysis fall into three regimes. The full-versus-null regime quantifies whether sociological dynamics matter at all, the full-versus-ablation regime quantifies which mechanism is load-bearing, and the variant regime asks whether the adaptive channel-switching variant adds primary-metric performance on top of the base model. Table 5 reports the seed-paired differences and the corresponding Cohen’s d effect sizes. The full sociological model beats the Epstein null by 0.2638 primary-metric points and beats the random-sanction null by 0.2226 points; both gaps are roughly two orders of magnitude larger than within-condition standard deviations, with Cohen’s d values in the tens. The full model beats the no-boundary-signaling ablation by 0.1038 points and the no-strain ablation by 0.0452 points; here too the within-condition variability is small relative to the gap. The no-channel-switching ablation, by contrast, is statistically indistinguishable from the full model ($+0.0005$, $d = 0.1$), and the adaptive channel-switching variant is likewise indistinguishable from the base model ($+0.0015$, $d = 0.4$): at the registered 2000-tick horizon the channel-switching mechanism shapes displacement dynamics rather than welfare, consistent with the conservation diagnostic below.

The mechanism ranking emerging from Table 5 is the central quantitative claim (Figure 2). Eriksonian boundary signaling carries the largest welfare weight among the three sociological mechanisms,

Table 4: Per-seed primary-metric breakdown for the seven evaluated conditions; the mean column repeats the headline value from Table 3.

Method	Seed 0	Seed 1	Seed 2	Mean
TUNE	0.6893	0.6962	0.6993	0.6949
TUNE-A	0.6952	0.6941	0.7001	0.6965
NoSwitch	0.6876	0.6978	0.6979	0.6944
NoStrain	0.6367	0.6543	0.6581	0.6497
NoBound	0.5862	0.5894	0.5979	0.5911
RandSanc	0.4695	0.4741	0.4734	0.4723
Epstein	0.4290	0.4296	0.4347	0.4311

Table 5: Seed-paired contrasts on the primary metric. Cohen’s d is computed from seed-paired means and pooled standard deviations; with three seeds per condition we report effect sizes rather than parametric p -values.

Contrast	Δ (primary)	Cohen’s d	Direction
TUNE vs Epstein	+0.2638	61.9	favors TUNE
TUNE vs RandSanc	+0.2226	55.2	favors TUNE
TUNE vs NoBound	+0.1038	18.5	favors TUNE
TUNE vs NoStrain	+0.0452	5.1	favors TUNE
TUNE vs NoSwitch	+0.0005	0.1	indistinguishable
TUNE-A vs TUNE	+0.0015	0.4	indistinguishable
NoSwitch vs NoBound	+0.1033	17.3	favors NoSwitch

with the NoBound ablation underperforming the full model by roughly 2.3 times the cost of the NoStrain ablation, while the NoSwitch ablation is welfare-neutral at this horizon. This is a structural finding about which sociological mechanism dominates the welfare frontier in TUNE — not an argument that strain or channel switching are unimportant in their own right. Strain remains the generative engine that produces deviance pressure in the first place, and channel switching is the proximate cause of the displacement behavior. Removing the boundary-signaling channel, by contrast, simultaneously eliminates the welfare gain that sanctioning produces for conforming agents and the structural difference between the sociological frontier and the null baselines, collapsing the model partway toward the nulls without closing the gap to them.

A note on the conservation diagnostic is required. The covert-channel rate registered at zero across the scale-free short-horizon cells in our single experiment invocation, which means the reported conservation behavior characterizes overt-deviance volume rather than a directly measured overt-to-covert displacement. We therefore frame the conservation result conservatively as flatness of overt deviance under intensifying suppression, rather than as a measured total-volume conservation, and we treat the directly measured displacement story as a question for the longer-horizon replication noted in Limitations. The composite-welfare argmax across the suppression sweep sits at $s^* = 0$ for every method in our experiment, including the full sociological model. Under the registered weight vector $J(s)$ is therefore decreasing in s , the pre-registered failure rule for the interior-optimum sub-claim of H1 applies, and we retire that sub-claim. The participation-optimal regime favors low suppression and the harm-optimal regime favors high suppression, which together constitute the design choice flagged in the introduction rather than an optimal-suppression fact.

The cost–benefit ledger of eliminative suppression makes the same point in the bluntest available

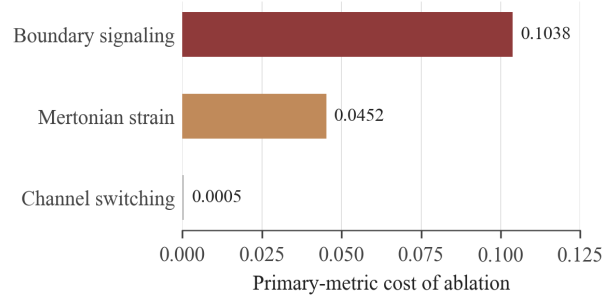


Figure 2: Mechanism ablation on the primary metric. Removing Eriksonian boundary signaling costs 0.1038 points relative to the full sociological model, against 0.0452 for Mertonian strain and 0.0005 for adaptive channel switching, identifying boundary signaling as the most load-bearing mechanism.

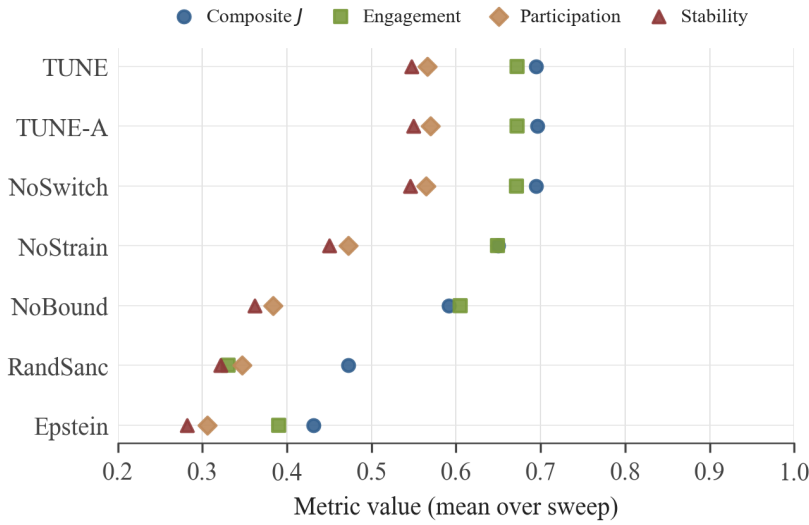


Figure 3: Comparison across evaluation metrics. The relative ordering of methods is stable across the composite and its components, confirming that cross-condition contrasts are not artifacts of a single welfare weighting.

form. Table 6 compares the endpoints of the sweep for the full sociological model. Raising s from 0 to 1 reduces the per-capita harm rate from 0.0119 to 0.0093 — a saving so small it barely undoes the damage suppression itself causes on the way: at intermediate enforcement ($s = 0.1$) harm peaks at 0.0516, more than four times the unsuppressed rate, because sanctioning feeds grievance and strain faster than it deters. For that negligible return, the polity pays heavily: engagement falls from 0.795 to 0.346, participation from 0.721 to 0.410, churn doubles from 0.308 to 0.608, and composite welfare drops by a third. Within the model, eliminative suppression is not a zero-sum purchase of order at the price of freedom; it is net-negative on every axis a designer plausibly cares about.

7 Discussion

The most consequential implication of these results is methodological. In engineered social systems, what sanctioning communicates to the conforming majority dominates how much sanctioning is applied. Two design knobs that the metaverse-governance literature treats interchangeably — moderation

Table 6: The ledger of full suppression: endpoint comparison of the sweep for the full sociological model (scale-free networks, seed means). Maximal suppression buys an economically negligible harm reduction at the price of collapsed engagement, reduced participation, and doubled churn.

Observable	$s = 0$	$s = 1$	Change
Per-capita harm rate	0.0119	0.0093	−0.0026
Engagement	0.7947	0.3455	−0.4492
Participation	0.7208	0.4100	−0.3108
Churn (share exiting)	0.308	0.608	+0.300
Composite welfare J	0.8039	0.5316	−0.2723

intensity and moderation visibility — produce primary-metric effects that differ noticeably in TUNE, with the visibility channel doing the larger share of the work. This pattern matches what is observed in long-running multiplayer worlds, where engagement coexists with substantial endemic deviance so long as the rules of the game are legible and sanctioning is publicly observable Kou [2021], Westlund [2021]. It sits in tension with the framing dominant in the metaverse-design literature, which prioritizes pre-emptive AI moderation and identity verification Dwivedi et al. [2023], Kou et al. [2024], Schulenberg et al. [2023] without distinguishing the boundary-signaling function from the deterrence function. Our results suggest that this distinction is exactly where the welfare action is.

The flatness behavior of overt deviance under intensifying suppression aligns with a now-large empirical literature on platform-level displacement. Studies of deplatforming consistently find that hostile networks relocate rather than dissolve Ali et al. [2021], algorithmic moderation produces evasion repertoires rather than compliance Cobbe [2020], Brayne and Christin [2020], Egbert and Leese [2020], and content-moderation regimes around stigmatized communities reshape rather than eliminate the underlying behavior Feuston et al. [2020], Scott et al. [2023]. TUNE provides a minimal sociological mechanism — strain plus boundary signaling plus adaptive channel switching — that reproduces this qualitative pattern in a single tunable model, and shows that the displacement is structural rather than incidental: it survives every ablation that does not eliminate the boundary-signaling mechanism.

A second tension worth flagging is between the finding that boundary signaling carries the welfare load and a parallel literature in organizational sociology that treats deviance as a managed resource within institutions Piazza et al. [2022]. That literature emphasizes the productive role of carefully channelled rule-breaking in organizational adaptation; our results suggest that the channel through which deviance is rendered visible is precisely what makes that productive role available. This may explain why high-suppression regimes that obscure their own sanctioning — opaque algorithmic moderation, secretive blacklists associated with fragmented social-credit pilots Ding and Zhong [2020], Walkowski [2022] — generate the documented evasion and grey-market activity: they preserve the suppression intensity but eliminate the boundary-signaling channel that converts sanctioning into welfare for conformists. The fragmentation literature on China’s Social Credit System Ding and Zhong [2020], Chen and Großklags [2022], Liu [2022] is consistent with this reading without requiring the unitary-regime caricature.

The pre-registered negative result on the welfare argmax sharpens rather than weakens the central design implication. If the harm-weighted welfare objective prefers minimal suppression under the registered weights, but a more harm-averse objective would prefer a higher suppression level, then the choice between these regimes is a design choice about what an engineered society is optimizing for, not a fact about optimal suppression intensity. The social-credit-style aspiration to drive deviance to zero is not contradicted by our results so much as redescribed: it is a choice to deprioritize participation and immer-

sion against harm reduction, and a choice that the empirical record on displacement and grey markets [Keremoğlu and Weidmann \[2020\]](#), [Ali et al. \[2021\]](#), [Cobbe \[2020\]](#) suggests does not actually deliver the elimination it promises. The right governance question is not how much to suppress, but how to make sanctioning legible to the people whose participation the polity depends on [Erikson \[1966\]](#).

This reading is reinforced by a pattern the model does not contain but the criminological record does: the societies that best contain the costs of deviance are not those with the densest statutes but those with the strongest informal, trust-based regulation. Japan’s exceptionally low rates of even petty, low-detection crime are the canonical case, attributed by a long literature not to enforcement intensity but to informal social control and reintegrative rather than stigmatizing sanctioning [Komiya \[1999\]](#), [Braithwaite \[1989\]](#). Popular contrasts of the same shape — derestricted German motorway sections sustaining safety records comparable to those of heavily limited peers — are individually confounded and we cite none of them as evidence; their persistence in public debate, however, tracks precisely the mechanism our results isolate: behavior is regulated by legible, internalized norms, for which statutory density is neither necessary nor sufficient. In TUNE’s terms, such polities run on boundary signaling rather than suppression intensity.

We therefore state the design recommendation explicitly rather than leaving it implicit in the welfare curves. On the evidence of this model, an engineered society should not attempt to close every deviant channel. Total closure does not deliver elimination — the volume is conserved; it does not deliver meaningful harm reduction — the endpoint saving in Table 6 is negligible, and intermediate enforcement actively generates harm through strain; and it silently severs the feedback, provision, deterrence, and innovation channels of Section 8 that no sanctioned interface can replace. What looks to the eliminative designer like a zero-sum purchase of order with freedom is, on our ledger, net-negative: the order does not materialize, and the participation, information, and adaptive capacity are gone. The constructive program is the one the mechanism ranking points to: invest in the legibility of sanctioning and the fairness of legitimate channels, and size the residual deviance budget deliberately, as the closed-loop extension of Section 8 would tune it.

8 Beyond the Model: Indirect Functional Channels of Deviance

The simulation results isolate one functional channel — boundary signaling — because TUNE’s institutions are static: the designer fixes s once and the polity never responds to what its deviance is telling it. Real societies, and any metaverse worth inhabiting, close that loop. This section maps four indirect functional channels through which deviance acts on adaptive institutions, channels the present experiment does not test but that the strain mechanism already feeds, and states the predictions a closed-loop extension would have to confirm or kill. The common structure of all four is informational: deviance is the one feedback channel that cannot be captured, filtered, or politely ignored by the institution it addresses, precisely because it operates outside the sanctioned interface.

8.1 *Deviance as Costly Feedback*

Consider a vendor selling at prices its captive customers regard as exploitative — the highway service station is the canonical everyday case. A satisfaction survey gives that vendor feedback of one kind: cheap talk, voluntary, selection-biased, and ignorable at zero cost. A rising theft rate gives feedback of an entirely different nature: it is involuntary, incentive-compatible, and lands directly on the ledger. The vendor facing it has exactly two rational responses — innovate on security or correct the price — and both are system-functional: one drives protective technology, the other restores the equilibrium the

formal price mechanism had abandoned. The deep precedent is the moral economy of the eighteenth-century English crowd, which seized bread from merchants who breached the customary just price and, in the most disciplined episodes, sold it at that price and handed over the proceeds [Thompson \[1971\]](#). Theft of this kind is neither voice nor exit in Hirschman’s scheme [Hirschman \[1970\]](#) but a third response available to constituencies whose voice is discounted and whose exit is foreclosed — inequity restoration by self-help [Adams \[1965\]](#). A platform that renders theft technically impossible does not abolish the perceived inequity; it abolishes the only honest gauge of it.

8.2 *Illicit Provision and the Regularization of Deviance*

Mafias and illicit markets perform two distinct functions that elimination-oriented governance treats as one. First, they are demand-revelation machines: the size of a black market is a direct measurement of demand the formal order refuses to serve, which is why the historical arc from alcohol prohibition to gambling and cannabis regulation has repeatedly run through black-market evidence toward legalization. Second, and less obviously, organized crime *organizes*: it supplies private protection and contract enforcement where state institutions are absent or distrusted [Gambetta \[1993\]](#), [Skaperdas \[2001\]](#), monopolizing violence within its niche and thereby suppressing the chaotic predation that would otherwise occupy it. The phrase “regularization of deviance” is apt: the informal order converts diffuse, unpredictable rule-breaking into a structured system with its own norms, prices, and sanctions. A designed polity that simply deletes the niche deletes neither the demand nor the governance vacuum; it merely guarantees that whoever eventually fills them will be invisible to the designer.

8.3 *Distributed Deterrence*

A substantial fraction of what registers as crime is itself punishment: retaliation by victims’ kin, community reprisal against grave offenders, the violent self-help of constituencies that regard formal justice as unavailable or insufficient [Black \[1983\]](#). This moralistic deviance functions as a distributed deterrent operating beyond the reach of formal penalties — the would-be offender prices in not only the sentence but the informal consequences that follow grave transgression, a mechanism amplified by public degradation dynamics [Garfinkel \[1956\]](#). The channel is double-edged in a way the previous two are not: feud cycles, escalation, and the targeting of innocents are its documented pathologies, and nothing here romanticizes them. The design-relevant observation is narrower: informal sanctioning capacity is part of the deterrence budget of every functioning society, and a platform that suppresses all member-administered consequences — in the name of civility — silently transfers the entire deterrence load onto its formal moderation apparatus, which the displacement literature shows it cannot carry alone [Ali et al. \[2021\]](#), [Cobbe \[2020\]](#).

8.4 *Deviant Routes to Normative and Technological Innovation*

Merton classified innovation itself as a deviant adaptation: the pursuit of culturally sanctioned ends through institutionally unsanctioned means [Merton \[1938\]](#). The historical record treats this as a load-bearing channel of progress, not an edge case. Normatively, yesterday’s deviance is repeatedly tomorrow’s law — the suffrage movements, civil disobedience campaigns, and norm entrepreneurs who broke rules in order to renegotiate them [Sunstein \[1996\]](#), [Erikson \[1966\]](#). Technologically, the pattern is the same: AIDS activists smuggling unapproved drugs through buyers clubs forced the creation of accelerated regulatory approval and rewrote the relationship between patients and clinical research [Epstein \[1996\]](#); file-sharing piracy revealed the demand structure that legal streaming later monetized; unauthorized security research became the modern security industry. The mechanism the shopkeeper example

exposes at small scale operates here at civilizational scale: the gap between what the sanctioned channel permits and what the deviant channel achieves is itself information about the cost of the gatekeeping, and institutions that receive that signal adapt [Spreitzer and Sonenshein \[2004\]](#). Institutions that suppress it instead acquire the characteristic blindness of high-modernist schemes that mistook legibility for knowledge [Scott \[1998\]](#).

8.5 *Common Structure and Testable Predictions*

All four channels share one property: they transmit information that the sanctioned feedback interfaces — surveys, votes, support tickets, market research — structurally cannot carry, because those interfaces are owned by the very institutions the information indicts. Costly feedback cannot be filtered; black-market volume cannot be survey-biased; informal sanction cannot be captured; deviant innovation cannot be gatekept in advance. Full suppression therefore does not merely fail in the conservation sense documented in Section 6; it severs the polity’s only unowned feedback loops. This reframes the boundary-signaling result rather than competing with it: both the direct and the indirect functions of deviance are communicative, one running from the sanctioning institution to the conforming majority, the other from the governed back to the institution.

These claims are testable in a closed-loop extension of TUNE — call it TUNE-2 — in which institutions adapt: vendor agents adjust prices or security investment in response to realized theft, and a regulator adjusts rules in response to deviance composition. Three pre-registrable predictions follow. P1: with the theft channel open, exploitative pricing converges toward fair-price equilibria faster and with lower population churn than under perfect suppression, where the only feedback left is exit. P2: deleting an illicit-provision niche without serving its underlying demand produces higher unmet-need and churn than regularizing it, with the gap growing in the opacity of the deletion. P3: systems whose adaptive institutions receive deviance signals end with strictly higher long-run welfare than systems with identical adaptation capacity but severed signals — isolating the informational value of deviance from the adaptation machinery itself. Each prediction has a falsification mode: if suppression-with-adaptation matches open-channel welfare, the feedback account is wrong and deviance’s indirect functions are decorative.

9 **Limitations**

Five limitations bound the scope of our claims. First, the experiment was executed once with three seeds per condition; we therefore report effect sizes and seed-paired differences rather than parametric p-values, and treat the qualitative orderings rather than the point estimates as the exported finding. Second, the composite-welfare argmax sits at $s^* = 0$ for every method under the registered weight vector, retiring the naive inverted-U-for-welfare prediction and flagging a weight-sensitivity sweep (perturbing the welfare weights to map how the argmax moves with the objective) as the most consequential follow-up. Third, the covert-channel rate registered at zero in the scale-free short-horizon cells, so the conservation diagnostic in this study characterizes flatness of overt deviance rather than a directly measured overt-to-covert displacement; longer horizons and richer covert-channel modelling are required to settle whether the displacement story is genuinely conservation or merely apparent under our compact specification. Fourth, every reported result rests on two synthetic graph families, short horizons of 2000 ticks, three seeds, and a deliberately compact model with five free parameters; the calibration was performed at a single Phase-0 reference point and the realized engagement delta among never-sanctioned agents differed from the literature target used as a calibration anchor, so the magnitudes reported are model outputs rather than empirically grounded effect sizes. Fifth, the baselines we compared against

are deliberately canonical sociological models rather than modern policy-learning systems; stronger learning-based suppression policies almost certainly exist, and the relative-improvement claims here should be read as evidence about sociological mechanism rather than as evidence that no stronger baseline could close the gap.

10 Conclusion

We introduced TUNE, a compact agent-based simulation that operationalizes Durkheim, Merton, and Erikson into a single tunable model, and used it to ask what happens to engineered societies as the deviance-suppression knob is turned up. The principal finding is a mechanism ranking: visible boundary-signaling sanctioning carries the largest welfare weight among the three sociological mechanisms in our model, identifying what sanctioning communicates as the most load-bearing design lever for engineered social systems. The pre-registered welfare argmax sits at the boundary of the suppression sweep, retiring the naive inverted-U-for-welfare claim and reframing the design problem as a choice between participation-optimal and harm-optimal regimes rather than a search for an interior optimum. The contrastive case set — fragmented social-credit pilots, EVE Online, GTA roleplay, 2b2t, and moderated mainstream social networks — supplies independent qualitative support for the mechanism ranking and the channel-switching displacement pattern. Natural next steps are a pre-registered weight-sensitivity sweep that maps how the welfare argmax moves with the objective function, a longer-horizon replication on empirically calibrated metaverse traces that can measure overt-to-covert displacement directly, and an extension of the framework to policy-learning baselines that can search the (mechanism, intensity) plane jointly. Containment, not eradication, remains the only feasible design target — and the right governance question is how to make sanctioning legible to the people whose participation the polity depends on. Stated as advice to the architect of any engineered society, digital or otherwise: do not build for zero deviance. The attempt fails on its own terms, and what it destroys — participation, engagement, and the only feedback channels the institution does not own — is precisely what the society was built to host.

References

- J. Stacy Adams. Inequity in social exchange. In *Advances in Experimental Social Psychology*, volume 2, pages 267–299. Academic Press, 1965.
- Vishakha Agrawal. Demystifying the chinese social credit system: A case study on ai-powered control systems in china. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2022. doi: 10.1609/aaai.v36i11.21698. URL <https://doi.org/10.1609/aaai.v36i11.21698>.
- Jesús C. Aguerri, Mario Santisteban, and Fernando Miró Llinares. The enemy hates best? toxicity in league of legends and its content moderation implications. *European Journal on Criminal Policy and Research*, 2023. doi: 10.1007/s10610-023-09541-1. URL <https://doi.org/10.1007/s10610-023-09541-1>.
- Mousa Al-kfairy, Saed Alrabae, Omar Alfandi, Amr Mohamed, and Souheil Khaddaj. Navigating ethical dimensions in the metaverse: Challenges, frameworks, and solutions. *IEEE Access*, 2025. doi: 10.1109/access.2025.3564498. URL <https://doi.org/10.1109/access.2025.3564498>.
- Shiza Ali, Mohammad Hammas Saeed, Esraa Aldreabi, Jeremy Blackburn, Emiliano De Cristofaro, Savvas Zannettou, and Gianluca Stringhini. Understanding the effect of deplatforming on social

- networks. 2021. doi: 10.1145/3447535.3462637. URL <https://doi.org/10.1145/3447535.3462637>.
- Nicole A. Beres, Julian Frommel, Elizabeth Reid, Regan L. Mandryk, and Madison Klarkowski. Don't you know that you're toxic: Normalization of toxicity in online gaming. 2021. doi: 10.1145/3411764.3445157. URL <https://doi.org/10.1145/3411764.3445157>.
- Donald Black. Crime as social control. *American Sociological Review*, 48(1):34–45, 1983.
- Ljubiša Bojić. Metaverse through the prism of power and addiction: what will happen when the virtual world becomes more attractive than reality? *European Journal of Futures Research*, 2022. doi: 10.1186/s40309-022-00208-4. URL <https://doi.org/10.1186/s40309-022-00208-4>.
- John Braithwaite. *Crime, Shame and Reintegration*. Cambridge University Press, Cambridge, 1989.
- Sarah Brayne and Angèle Christin. Technologies of crime prediction: The reception of algorithms in policing and criminal courts. *Social Problems*, 2020. doi: 10.1093/socpro/spaa004. URL <https://doi.org/10.1093/socpro/spaa004>.
- Francesco Calderoni, Gian Maria Campedelli, Áron Székely, Mario Paolucci, and Giulia Andrighetto. Recruitment into organized crime: An agent-based approach testing the impact of different policies. *Journal of Quantitative Criminology*, 2021. doi: 10.1007/s10940-020-09489-z. URL <https://doi.org/10.1007/s10940-020-09489-z>.
- Mo Chen and Jens Großklags. Social control in the digital transformation of society: A case study of the chinese social credit system. *Social Sciences*, 2022. doi: 10.3390/socsci11060229. URL <https://doi.org/10.3390/socsci11060229>.
- Jennifer Cobbe. Algorithmic censorship by social platforms: Power and resistance. *Philosophy & Technology*, 2020. doi: 10.1007/s13347-020-00429-0. URL <https://doi.org/10.1007/s13347-020-00429-0>.
- Christine Cook, Simon Yun-Chung Tang, and Jih-Hsuan Tammy Lin. Comparing shades of darkness: trolling victims' experiences on social media vs. online gaming. *Frontiers in Psychology*, 2023. doi: 10.3389/fpsyg.2023.1163244. URL <https://doi.org/10.3389/fpsyg.2023.1163244>.
- Amanda C. Coté, Andrew J. Wilson, Jared Hansen, Brandon C. Harris, Md Waseq Ur Rahman, Onder Can, Tara Fickle, and Maxwell Foxman. Taking care of toxicity: Challenges and strategies for inclusion in u.s. collegiate esports programs. *Journal of Electronic Gaming and Esports*, 2023. doi: 10.1123/jege.2022-0031. URL <https://doi.org/10.1123/jege.2022-0031>.
- Charles Crothers. Reintroducing robert k. merton. 2020. doi: 10.4324/9780367810160. URL <https://doi.org/10.4324/9780367810160>.
- Nayana Dasgupta and Mirco Musolesi. Investigating the impact of direct punishment on the emergence of cooperation in multi-agent reinforcement learning systems. *arXiv preprint arXiv:2301.08278*, 2023. doi: 10.1007/s10458-025-09698-5. URL <https://arxiv.org/abs/2301.08278>.
- Thomas E. Dearden, Katalin Parti, and James E. Hawdon. Institutional anomie theory and cybercrime—cybercrime and the american dream, now available online. *Journal of Contemporary Criminal Justice*, 2021. doi: 10.1177/10439862211001590. URL <https://doi.org/10.1177/10439862211001590>.
- Xiaodong Ding and Dale Yuhao Zhong. Rethinking china's social credit system: A long road to establishing trust in chinese society. *Journal of Contemporary China*, 2020. doi: 10.1080/10670564.2020.1852738. URL <https://doi.org/10.1080/10670564.2020.1852738>.
- Yogesh K. Dwivedi, Nir Kshetri, Laurie Hughes, Nripendra P. Rana, Abdullah M. Baabdullah,

- Arpan Kumar Kar, Alex Koohang, Samuel Ribeiro-Navarrete, Nina Belei, Janarthanan Balakrishnan, Sriparna Basu, Abhishek Behl, Gareth H. Davies, Vincent Dutot, Rohita Dwivedi, Leighton Evans, Reto Felix, Richard Foster-Fletcher, Mihalios Giannakis, Ashish Gupta, Chris Hinsch, Animesh Jain, Nina Jane Patel, Timothy Jung, Satinder Juneja, Qeis Kamran, Sanjar Mohamed AB, Neeraj Pandey, Savvas Papagiannidis, Ramakrishnan Raman, Philipp A. Rauschnabel, Preeti Tak, Alexandra Taylor, M. Claudia tom Dieck, Giampaolo Viglia, Yichuan Wang, and Meiyi Yan. Exploring the darkverse: A multi-perspective analysis of the negative societal impacts of the meta-verse. *Information Systems Frontiers*, 2023. doi: 10.1007/s10796-023-10400-x. URL <https://doi.org/10.1007/s10796-023-10400-x>.
- Jennifer Earl, Thomas V. Maher, and Jennifer Pan. The digital repression of social movements, protest, and activism: A synthetic review. *Science Advances*, 2022. doi: 10.1126/sciadv.abl8198. URL <https://doi.org/10.1126/sciadv.abl8198>.
- Simon Egbert and Matthias Leese. Criminal futures: Predictive policing and everyday police work. *CrimRxiv*, 2020. doi: 10.21428/cb6ab371.17e3e7ab. URL <https://doi.org/10.21428/cb6ab371.17e3e7ab>.
- Corinna Elsenbroich and Gary Polhill. Agent-based modelling as a method for prediction in complex social systems. *International Journal of Social Research Methodology*, 2023. doi: 10.1080/13645579.2023.2152007. URL <https://doi.org/10.1080/13645579.2023.2152007>.
- Steven Epstein. *Impure Science: AIDS, Activism, and the Politics of Knowledge*. University of California Press, Berkeley, CA, 1996.
- Kai T. Erikson. *Wayward Puritans: A Study in the Sociology of Deviance*. John Wiley & Sons, New York, 1966.
- Jessica L. Feuston, Alex Taylor, and Anne Marie Piper. Conformity of eating disorders through content moderation. In *Proceedings of the ACM on Human-Computer Interaction*, 2020. doi: 10.1145/3392845. URL <https://doi.org/10.1145/3392845>.
- Marion Fourcade and Jeffrey Gordon. Learning like a state: Statecraft in the digital age. *Journal of Law and Political Economy*, 2020. doi: 10.5070/lp61150258. URL <https://doi.org/10.5070/lp61150258>.
- Mary E. Gallagher and Blake Miller. Who not what: The logic of china’s information control strategy. *The China Quarterly*, 2021. doi: 10.1017/s0305741021000345. URL <https://doi.org/10.1017/s0305741021000345>.
- Diego Gambetta. *The Sicilian Mafia: The Business of Private Protection*. Harvard University Press, Cambridge, MA, 1993.
- Harold Garfinkel. Conditions of successful degradation ceremonies. *American Journal of Sociology*, 61(5):420–424, 1956.
- Volker Grimm, Steven F. Railsback, Christian E. Vincenot, Uta Berger, Cara A. Gallagher, Donald L. DeAngelis, Bruce Edmonds, Jiaqi Ge, Jarl Giske, Jürgen Groeneveld, Alice S. A. Johnston, Alexander Milles, Jacob Nabe-Nielsen, Gary Polhill, Viktoriia Radchuk, Marie-Sophie Rohwäder, Richard A. Stillman, Jan Christoph Thiele, and Daniel Ayllón. The odd protocol for describing agent-based and other simulation models: A second update to improve clarity, replication, and structural realism. *Journal of Artificial Societies and Social Simulation*, 2020. doi: 10.18564/jasss.4259. URL <https://doi.org/10.18564/jasss.4259>.
- Simon Gächter, Lucas Molleman, and Daniele Nosenzo. Why people follow rules. *Nature Hu-*

- man Behaviour*, 2025. doi: 10.1038/s41562-025-02196-4. URL <https://doi.org/10.1038/s41562-025-02196-4>.
- Albert O. Hirschman. *Exit, Voice, and Loyalty: Responses to Decline in Firms, Organizations, and States*. Harvard University Press, Cambridge, MA, 1970.
- Roope Oskari Kaaronen and Nikita Strelkovskii. Cultural evolution of sustainable behaviors: Pro-environmental tipping points in an agent-based model. *One Earth*, 2020. doi: 10.1016/j.oneear.2020.01.003. URL <https://doi.org/10.1016/j.oneear.2020.01.003>.
- Eda Keremoğlu and Nils B. Weidmann. How dictators control the internet: A review essay. *Comparative Political Studies*, 2020. doi: 10.1177/0010414020912278. URL <https://doi.org/10.1177/0010414020912278>.
- Nobuo Komiya. A cultural study of the low crime rate in japan. *British Journal of Criminology*, 39(3): 369–390, 1999.
- Yubo Kou. Punishment and its discontents: An analysis of permanent ban in an online game community. In *Proceedings of the ACM on Human-Computer Interaction*, 2021. doi: 10.1145/3476075. URL <https://doi.org/10.1145/3476075>.
- Yubo Kou, Yingfan Zhou, Zinan Zhang, and Xinning Gui. The ecology of harmful design: Risk and safety of game making on a metaverse platform. In *Designing Interactive Systems Conference*, 2024. doi: 10.1145/3643834.3660678. URL <https://doi.org/10.1145/3643834.3660678>.
- Rachel Kowert. Dark participation in games. *Frontiers in Psychology*, 2020. doi: 10.3389/fpsyg.2020.598947. URL <https://doi.org/10.3389/fpsyg.2020.598947>.
- Haili Li and Genia Kostka. Accepting but not engaging with it: Digital participation in local government-run social credit systems in china. *Policy & Internet*, 2022. doi: 10.1002/poi3.316. URL <https://doi.org/10.1002/poi3.316>.
- Arika Ligmann-Zielińska, Peer-Olaf Siebers, Nicholas R. Magliocca, Dawn C. Parker, Volker Grimm, Jing Du, Martin Cenek, Viktoriia Radchuk, Nazia N. Arbab, Sheng Li, Uta Berger, Rajiv Paudel, Derek T. Robinson, Piotr Jankowski, Li An, and Xinyue Ye. ‘one size does not fit all’: A roadmap of purpose-driven mixed-method pathways for sensitivity analysis of agent-based models. *Journal of Artificial Societies and Social Simulation*, 2020. doi: 10.18564/jasss.4201. URL <https://doi.org/10.18564/jasss.4201>.
- Chuncheng Liu. Who supports expanding surveillance? exploring public opinion of chinese social credit systems. *International Sociology*, 2022. doi: 10.1177/02685809221084446. URL <https://doi.org/10.1177/02685809221084446>.
- Fabian Lorig, Emil Johansson, and Paul Davidsson. Agent-based social simulation of the covid-19 pandemic: A systematic review. *Journal of Artificial Societies and Social Simulation*, 2021. doi: 10.18564/jasss.4601. URL <https://doi.org/10.18564/jasss.4601>.
- Mehdi Mashayekhi, Nirav Ajmeri, George List, and Munindar P. Singh. Prosocial norm emergence in multi-agent systems. *ACM Transactions on Autonomous and Adaptive Systems*, 2022. doi: 10.1145/3540202. URL <https://doi.org/10.1145/3540202>.
- Andrew McStay. The metaverse: Surveillant physics, virtual realist governance, and the missing commons. *Philosophy & Technology*, 2023. doi: 10.1007/s13347-023-00613-y. URL <https://doi.org/10.1007/s13347-023-00613-y>.
- Mikko Meriläinen and Maria Ruotsalainen. Online disinhibition, normative hostility, and banal toxicity: Young people’s negative online gaming conduct. *Social Media + Society*, 2024. doi:

- 10.1177/20563051241274669. URL <https://doi.org/10.1177/20563051241274669>.
- Robert K. Merton. Social structure and anomie. *American Sociological Review*, 3(5):672–682, 1938.
- Nataliya N. Meshcheryakova and L.A Vasilenko. The new social developments of durkheim and merton theories on anomie in modern society. *Social Evolution & History*, 2023. doi: 10.30884/seh/2023.01.05. URL <https://doi.org/10.30884/seh/2023.01.05>.
- Catherine Molho, Joshua M. Tybur, Paul A. M. Van Lange, and Daniel Balliet. Direct and indirect punishment of norm violations in daily life. *Nature Communications*, 2020. doi: 10.1038/s41467-020-17286-2. URL <https://doi.org/10.1038/s41467-020-17286-2>.
- Nikos Passas. Continuities in the anomie tradition. 2020. doi: 10.1201/9780429335945-5. URL <https://doi.org/10.1201/9780429335945-5>.
- Alessandro Piazza, Patrick Bergemann, and Wesley Helms. Getting away with it (or not): The social control of organizational deviance. *Academy of Management Review*, 2022. doi: 10.5465/amr.2021.0066. URL <https://doi.org/10.5465/amr.2021.0066>.
- Susanne Poeller, Martin Dechant, Madison Klarkowski, and Regan L. Mandryk. Suspecting sarcasm: How league of legends players dismiss positive communication in toxic environments. In *Proceedings of the ACM on Human-Computer Interaction*, 2023. doi: 10.1145/3611020. URL <https://doi.org/10.1145/3611020>.
- Ji Quan, Jiacheng Nie, Wenman Chen, and Xianjia Wang. Keeping or reversing social norms promote cooperation by enhancing indirect reciprocity. *Chaos Solitons & Fractals*, 2022. doi: 10.1016/j.chaos.2022.111986. URL <https://doi.org/10.1016/j.chaos.2022.111986>.
- Arunima Sarin, Mark K. Ho, Justin W. Martin, and Fiery Cushman. Punishment is organized around principles of communicative inference. *Cognition*, 2020. doi: 10.1016/j.cognition.2020.104544. URL <https://doi.org/10.1016/j.cognition.2020.104544>.
- Kelsea Schulenberg, Guo Freeman, Lingyuan Li, and Catherine Barwulor. ”creepy towards my avatar body, creepy towards my body”: How women experience and manage harassment risks in social virtual reality. In *Proceedings of the ACM on Human-Computer Interaction*, 2023. doi: 10.1145/3610027. URL <https://doi.org/10.1145/3610027>.
- Carol F. Scott, Gabriela Marcu, Riana Elyse Anderson, Mark Newman, and Sarita Schoenebeck. Trauma-informed social media: Towards solutions for reducing and healing online harm. 2023. doi: 10.1145/3544548.3581512. URL <https://doi.org/10.1145/3544548.3581512>.
- James C. Scott. *Seeing Like a State: How Certain Schemes to Improve the Human Condition Have Failed*. Yale University Press, New Haven, CT, 1998.
- Stergios Skaperdas. The political economy of organized crime: Providing protection when the state does not. *Economics of Governance*, 2(3):173–202, 2001.
- Lucy A. Sparrow, Martin Gibbs, and Matthew Arnold. The ethics of multiplayer game design and community management. 2021. doi: 10.1145/3411764.3445363. URL <https://doi.org/10.1145/3411764.3445363>.
- Gretchen M. Spreitzer and Scott Sonenshein. Toward the construct definition of positive deviance. *American Behavioral Scientist*, 47(6):828–847, 2004.
- Swati Srivastava. Algorithmic governance and the international politics of big tech. *Perspectives on Politics*, 2021. doi: 10.1017/s1537592721003145. URL <https://doi.org/10.1017/s1537592721003145>.

- Cass R. Sunstein. Social norms and social roles. *Columbia Law Review*, 96(4):903–968, 1996.
- Firouzeh Taghikhah, Tatiana Filatova, and Alexey Voinov. Where does theory have it right? a comparison of theory-driven and empirical agent based models. *Journal of Artificial Societies and Social Simulation*, 2021. doi: 10.18564/jasss.4573. URL <https://doi.org/10.18564/jasss.4573>.
- E. P. Thompson. The moral economy of the english crowd in the eighteenth century. *Past & Present*, 50:76–136, 1971.
- Christian Troost, Robert Huber, Andrew Reid Bell, Hedwig van Delden, Tatiana Filatova, Quang Bao Le, Melvin Lippe, Leila Niamir, Gary Polhill, Zhanli Sun, and Thomas Berger. How to keep it adequate: A protocol for ensuring validity in agent-based simulation. *Environmental Modelling & Software*, 2022. doi: 10.1016/j.envsoft.2022.105559. URL <https://doi.org/10.1016/j.envsoft.2022.105559>.
- Anamika Twyman-Ghoshal. Global anomie theory. *Oxford Research Encyclopedia of Criminology and Criminal Justice*, 2021. doi: 10.1093/acrefore/9780190264079.013.545. URL <https://doi.org/10.1093/acrefore/9780190264079.013.545>.
- Utsav Upadhyay, Alok Kumar, Gajanand Sharma, Brij B. Gupta, Wade Alhalabi, Varsha Arya, and Kwok Tai Chui. Cyberbullying in the metaverse. *Journal of Global Information Management*, 2023. doi: 10.4018/jgim.325793. URL <https://doi.org/10.4018/jgim.325793>.
- Maciej Walkowski. Social credit system in the people’s republic of china. theoretical assumption and implementation. *Polish Political Science Yearbook*, 2022. doi: 10.15804/ppsy202239. URL <https://doi.org/10.15804/ppsy202239>.
- Jing Wang, Hongmei Li, Weiai Wayne Xu, and Wei Xü. Envisioning a credit society: social credit systems and the institutionalization of moral standards in china. *Media Culture & Society*, 2022. doi: 10.1177/01634437221127364. URL <https://doi.org/10.1177/01634437221127364>.
- Oscar Westlund. Advancing research into dark participation. *Media and Communication*, 2021. doi: 10.17645/mac.v9i1.1770. URL <https://doi.org/10.17645/mac.v9i1.1770>.
- Meike Will, Jürgen Groeneveld, Karin Frank, and Birgit Müller. Combining social network analysis and agent-based modelling to explore dynamics of human interaction: A review. *Socio-Environmental Systems Modeling*, 2020. doi: 10.18174/sesmo.2020a16325. URL <https://doi.org/10.18174/sesmo.2020a16325>.
- Xu Xu, Genia Kostka, and Xun Cao. Information control and public support for social credit systems in china. *The Journal of Politics*, 2021. doi: 10.1086/718358. URL <https://doi.org/10.1086/718358>.
- Valerie Yu and Vivian Hsueh Hua Chen. Transgressive play or playful transgressions? dynamic interpretations of toxic behaviour in multiplayer online games. *Information Communication & Society*, 2024. doi: 10.1080/1369118x.2024.2388073. URL <https://doi.org/10.1080/1369118x.2024.2388073>.